

Personal Profile



Dr. Mahesh Kumar

Department of Electrical Engineering, Mehran University of Engineering and
Technology Jamshoro, Sindh, Pakistan

Professional Engineer, (Pak. Engineering Council),

PhD., Universiti Teknologi Petronas, Malaysia

MEng. & BEng. Mehran University of Eng & Tech. Pakistan

Senior Member, Institute of Electrical and Electronics Engineering (IEEE),

Founder and Chair IEEE Power Electronics Society (PELS) Karachi Section,

Secretary, IEEE Power and Energy Society (PES) Karachi Section

Approved **PHD supervisor** (Higher Education Commission, Pak)

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Professional Summary

Dr. Mahesh Kumar is a Professional Engineer/Academician in Electrical Engineering with over 17 years of hands-on experience in teaching, research, and addressing industry-led engineering problem solving. Holds a PhD in Electrical and Electronics Engineering from Universiti Teknologi Petronas, Malaysia and has published more than 50 peer-reviewed articles, garnering 1,350+ citations and a combined impact factor of over 80+. His work focuses on renewable energy integration, battery energy storage, electric vehicle, load management integrated with artificial intelligence (computational intelligence, i.e. multiobjective particle swarm optimization), converters, WECS, and resilient power distribution networks under grid-connected and islanded conditions.

Honors, Recognition, and Outstanding Achievements

- Associate Professor, Assistant Professor and Lecturer in Electrical Engineering Department at Mehran University of Engineering and Technology Jamshoro.
- Received Graduate Assistantship (GA) for 4-years from Universiti Teknologi Petronas, Malaysia for pursuing PHD research.
- Received the Doctor of Philosophy (PhD.) in Electrical and Electronics Engineering Universiti Teknologi Petronas, Malaysia.
- Master in Engineering (M.Eng) in Electrical Power Engineering and Bachelors in Engineering (B. Eng) in Electrical Engineering from Mehran University of Engineering and Technology Jamshoro, Pakistan.
- Program Evaluator (PEV) for accreditation of Electrical Engineering Program from Pakistan Engineering Council.
- Professional Engineer, (P.Eng) registered with Pakistan Engineering Council.
- Reviewer of Higher Education commission of Pakistan Research Grants and Projects (reviewed million rupees research proposals).
- Senior Member: Institute of Electrical & Electronic Engineers USA, Founder and Chair IEEE Power Electronics (PELS) Karachi Section, Secretary, IEEE Power Energy Society (PES) Karachi Section and Advisor and Counselor at Mehran university of Engineering and Technology Jamshoro Pakistan.
- Served as Internal and External experts for undergrad, masters and PHD level.
- Author of 40 journal articles, 6 conference papers and 5 book chapters.
- Editorial Board Member of two journals, and key committee members of National and International Conferences.
- Received Outstanding Engineering Award and Excellence Award from IEEE Power Energy Society, and from IEEEPP respectively.

Professional Experience

Associate Professor

11-07-2019 to present

Department of Electrical Engineering, Mehran University of Engineering and Technology, Jamshoro, Sindh, Pakistan.

- Responsible for teaching major courses to undergraduate and postgraduate students, developing curriculum, and mentoring students. The role includes conducting research, publishing in reputable journals, supervising graduate students, and securing research funding. Responsibilities also involve contributing to academic governance, accreditation, quality assurance, and providing effective departmental and academic leadership.

Assistant Professor

28-05-2018 to 10-07-2019

Department of Electrical Engineering, Mehran University of Engineering and Technology, Jamshoro, Sindh, Pakistan.

Part-time Lecturer

June 2014- April 2018

Department of Electrical Engineering, Universiti Teknologi Petronas (UTP), Malaysia.

- Laboratory demonstration for undergrad courses such as Electrical Power System, Electrical Machines, etc. between the years 2014 – 2018.
- Helped FYP students at UTP for power loss minimization and voltage stability indexed improvement in RDS system.

Lecturer

16-08-2009 to 27-05-2018

Department of Electrical Engineering, Mehran University of Engineering and Technology, Jamshoro, Sindh, Pakistan

Lab Lecturer

31-01-2009 to 15-08-2009

Department of Electrical Engineering, Mehran University of Engineering and Technology, Jamshoro, Sindh, Pakistan

Academic Qualifications

Doctor of Philosophy (Ph.D.) in Electrical and Electronic Engineering, **July 2014- May 2018**
Universiti Teknologi Petronas, Malaysia.

- Optimal Integration of Renewable Distributed Generations in Radial Distribution System.
- The major contribution of the study is to offer optimal siting and sizing of intermittent and dispatchable distributed generation using multi-objective particle swarm optimization to reduce power losses, reduce voltage deviation and enhance voltage stability index in low voltage radial distribution network.

Master of Engineering (M.E.) in Electrical Power Engineering, Mehran University of Engineering and Technology, Jamshoro, Sindh, Pakistan. **Dec 2008 – April 2012**

- Energy Audit of Electrical Motors installed at AMZ Spinning Mill Kotri
- 82.88% with 3.75/4.0 CGPA. (16-04-2012)

Bachelor of Engineering (B.E.) in Department of Electrical Engineering, Mehran University of Engineering and Technology, Jamshoro, Sindh, Pakistan. **Dec 2003 –March 2008**

- 79.79% with 3.6/4.0 CGPA. (25-03-2008)

Professional License, Affiliation and Services

- **Professional Engineer** (Elect/23829) with Pakistan Engineering Council.
- **Founder and Chair**, IEEE Power Electronics Society, IEEE Karachi Section.
- **Secretary**, IEEE Power & Energy Society USA, Karachi Section
- **Senior Member**, Institution of Electrical and Electronic Engineering (IEEE), USA 94583638
- **Finance Secretary / Treasurer**, IEEEEP (Institution of Electrical & Electronics Engineers Pakistan), Local Centre Hyderabad (2022-2024)
- **Member**, Institution of Electrical and Electronics Engineers Pakistan (IEEEEP)

Professional Expertise

My professional expertise focuses on **Smart Grids, Microgrids, and Renewable Energy Systems**, with strong emphasis on **advanced control strategies, optimization techniques, and techno-economic analysis** for **sustainable power system planning**. I integrate **computational intelligence methods, centralized and decentralized control schemes, and large-scale optimization modeling** to address contemporary challenges in modern distribution networks. My work includes **optimal planning of intermittent and non-intermittent renewable energy sources, capacitor bank allocation for voltage stability improvement and loss reduction, and multi-objective optimization of distribution systems integrated with electric vehicle (EV) charging infrastructure**.

In addition, I have conducted research on **effective utilization of distributed energy resources (DERs) in islanded microgrids under power mismatch conditions**, as well as **optimal battery energy storage system (BESS) integration** in networks with **high renewable penetration**. My contributions extend to **techno-economic assessments of off-grid PV–wind–hydrogen-based EV charging stations across climatically diverse regions of Pakistan, on-grid solar PV feasibility analysis for industrial loads, and rural electrification through off-grid PV system design in Tharparkar, Pakistan**. I have also worked on **dynamic modeling and advanced control of DFIG-based wind energy systems, including Super-Twisting Fractional Order Terminal Sliding Mode Controllers, and on reactive power compensation and power quality enhancement in distribution networks**.



Mahesh Kumar

Mehran University of Engineering and Technology

Verified email at faculty.muett.edu.pk - [Homepage](#)

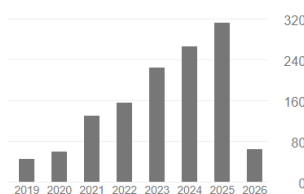
[Power system](#) [Renewable Energy](#) [Artificial intelligence](#)

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Co-authors

[EDIT](#)

	Amir Mahmood Soomro Mehran University of Engineerin...	>
	Perumal Nallagownden Universiti Teknologi PETRONAS	>
	Laveet Kumar, PhD Qatar University, Mehran Univer...	>
	Zubair Ahmed Memon Department of Electrical Enginee...	>
	Mansoor Soomro Assistant Professor, Department ...	>
	Ghulam Sarwar Kaloi	>

☐ TITLE	CITED BY	YEAR
☐ Social, Economic, and Environmental Impacts of M Kumar Wind solar hybrid renewable energy system 227	402	2020
☐ Advanced Pareto Front Non-Dominated Sorting Multi-Objective Particle Swarm Optimization for Optimal Placement and Sizing of Distributed Generation K Mahesh, P Nallagownden, I Elamvazuthi Energies 9 (12,982)	106	2016
☐ Optimal placement and sizing of renewable distributed generations and capacitor banks into radial distribution systems M Kumar, P Nallagownden, I Elamvazuthi Energies 10 (6), 811	78	2017
☐ Strategic planning of renewable distributed generation in radial distribution system using advanced MOPSO method MZ Malik, M Kumar, AM Soomro, MH Baloch, M Farhan, M Gul, GS Kaloi Energy Reports 6, 2872-2886	72	2020
☐ Optimal multi-objective placement and sizing of distributed generation in distribution system: a comprehensive review M Kumar, AM Soomro, W Uddin, L Kumar Energies 15 (21), 7850	67	2022
☐ Wind energy integration: Dynamic modeling and control of DFIG based on super twisting fractional order terminal sliding mode controller MA Soomro, ZA Memon, M Kumar, MH Baloch Energy Reports 7, 6031-6043	60	2021

Teaching & Learning

The courses taught to undergrad and postgrad along with few core and major subjects' insight.

Undergraduate Taught Courses		Postgraduate (ME/PHD) Taught Courses	
Power Distribution & Utilization	EL-425	Smart Grid Technologies and Application	ELP-632
Electrical Power Transmission	EL-323	Renewable Energy Integration and Optimization	ELP-643
Power System Analysis	EL-416	Wind Energy Conversion and Systems	EnSE-11
High Voltage Engineering	EL-424	Electrical Power Distribution Systems	ELP-626
Power System Stability and Control	EL-422	Electrical Power Engineering Lab	ELP-
Electronic Devices & Circuits	EL-211	Electrical Power Transmission	ELP-612
Digital Logic Design	EL-224		
Applied Physics	ES-112		
Applied Electricity (Th+Pr)	EL-111		

Smart Grid Technologies and Application

Elements and Technologies of Smart Grid System, Distributed Generation Resources, Introduction to energy storage devices, Different types of energy storage technologies, Analytical modelling of energy storage devices, Optimal sizing and siting of storages, Battery management system (BMS) Demand Response, Definition, Applications, Pricing and Energy Consumption Scheduling, Controllable Load Models, Dynamics, and Challenges, Electric Vehicles and Vehicle-to-Grid Systems, Demand Side Ancillary Services, Wide area Monitoring Systems, Sensor Networks, Phasor Measurement Units, Communications Infrastructure, Fault Detection and Self-Healing Systems, Applications and Challenges.

Wind Energy Conversion and System

Wind, its structure, statistics, measurements, data presentation, power in wind. Wind turbines –types, classifications, parameters. Wind turbine aerodynamics, Wind turbine loads, aerodynamic loads in steady operation, wind turbulence, static-dynamic fatigue analysis, yawed operation and tower shadow, Wind energy Conversion system (WECS) control system, requirements and strategies. WECS siting, rotor selection, Annual Energy output, Economics, and Social and environmental benefits/issues of WECS. Synchronous and asynchronous generators and loads, integration of wind energy converters to electrical networks, inverters. Testing of WECS

Renewable Energy Integration and Optimization

Concept of Renewable distributed generations (DGs), conventional and Renewable DGs, Centralized and Decentralized Power Generation and their Resources Modelling, Design and Evaluation of Standalone, Grid-Connected and Large-Scale renewable integrated power system, dispatchable and intermittent generation. Sizing and placement problems of these renewable generation with existing system. Converters, Power Converters in designing renewable DGs, their applications and modeling. Grid modeling, impact of high penetration of renewable generation on power system parameters, voltage dependent load models, load growth modeling, and EVs as future load. Power Flow Models - Modeling Tools and Solvers, Mixed-Integer D-OPF Model, Mixed-Integer Non-Linear Formulation, Mixed-Integer Convex Formulation, comparison of different power flow methods. Control Parameters- Technical (Power Losses, Voltage Deviation, Voltage Stability, and Reliability, overloading/under loading, power quality and protection coordination.) Economic and Environmental parameters. Optimizations- Linear approach, non –linear approach, Artificial Intelligence in grid connected distributed generation in distribution system.

Electrical Power Distribution Systems

Primary, secondary and Tertiary voltages. Radial and Ring Main systems. Estimation of load. Load characteristics. Substation-Switch gear and busbar arrangements. Calculation of voltage drop and regulation in distribution feeders. Power distribution voltages in Pakistan, causes, arcing on power distribution lines. Grounding and earthing, power factor, Importance of DC systems Main types

of batteries and their working, Battery charging, maintenance testing, Electroplating, Electrolysis, Electro metallurgical processes, Cathodic protection of poles, gas pipes and water structures.

Power System Stability and Control

Definition and classification of power system stability. Steady state and Transient Stability. The swing equation. Equal area criterion: application to a sudden increase in power input and application to a short circuit fault. Critical fault clearing time. Need and objective of power system control. Automated power system control concept, Power system operation modes. Supervisory control, Variables and time intervals for data acquisition, Monitoring and event processing, Control functions, Disturbance data collection and analysis, Human-Machine communication methods, active power and frequency control, Reactive power and voltage control, Economic dispatch,

Electrical Power Transmission

The course provides comprehensive coverage of AC and DC transmission systems, transmission line modeling (short, medium, and long lines), travelling waves, power flow analysis, and voltage regulation. It includes sag and tension calculations, corona effects, insulation systems, underground cables, EHV systems, and HVDC transmission technologies, along with protection, testing, and fault-locating techniques.

CONCEPT TO DESIGN AND DEVELOP SMART/SUPER GRID COURSEWORK

Smart grid, Renewable integration.

Training for Continuous Professional Development

1. Attended one day (1 CPD) webinar on “Service Leadership and Professional Ethics” on 13 May 2025, as continuing professional development program by Pakistan Engineering Council.
2. Attended one day (1 CPD) webinar on “The role of Engineers in safe and sustainable operations” on 16th February 2025 as continuing professional development program by Pakistan Engineering Council.
3. Attended one day (1 CPD) webinar on “Professional Ethics” on 30 January 2025 as continuing professional development program by Pakistan Engineering Council.
4. Attended one day (1 CPD) webinar on "The Future use Cases of BlockChain for Cyber Security" on 20 October 2024 as continuing professional development program by Pakistan Engineering Council.
5. Attended Half-day (0.5 CPD) webinar on “Leadership and Engineering: A Dichotomy or Misunderstanding” on 19th June 2021 initiated as continuing professional development program by Pakistan Engineering Council.
6. Attended Half-day (0.5 CPD) webinar on “Massive Connectivity in 5G+ Networks for Smart Cities” on 1st June 2021 initiated as continuing professional development program by Pakistan Engineering Council.

Higher Education Commission of Pakistan (HEC) Approved PHD Supervisor

- HEC Approved PhD Supervisor in the discipline of Engineering & Technology, from Higher Education Commission, Pakistan. Since 12-Jun-2019 till June 2025.

Achievement Awards

- Excellence Award by Institute of Electrical and Electronics Pakistan (IEEEEP) Hyderabad Local Centre at Mehran University, Jamshoro at one day session titled "MEET THE PERSONALITY" at Senate Hall of the University.
- Outstanding Engineer Award, IEEE Power and Energy Society Chapter For years of dedicated volunteer service and valuable contributions to IEEE PES Chapter and positively shaped the electric power and energy industry By Karachi Power and Energy Chapter, February 2024

Research/Travel Grants

- Higher Education Pakistan, Startup Research grant of Pkr. 369,000/- Optimal Integration of Renewable Distributed Generations in the Radial Distribution system.
- Ignite, National Technology Fund, Research and Development, Final Year Projects (FYP) Fund of Pkr. 70,000 /- for IOT based Smart Home Energy Management System
- IEEE – PES Humanitarian Technology Project Design and Implementation (HTPDI) 2023 project for Solar Energy Training and Panels for Flood- Affected Communities worth 2000USD.
- 1000 USD for opening and revitalizing IEEE PELS Karachi Section and won more than 1000+ USD for IEEE PES MUET Chapter.
- IEEE Power & Energy Society - Karachi Chapter has been awarded 970 USD by IEEE Power & Energy Society under 2024 IEEE PES High Performing Chapter Program (HPCP). An additional amount of 100 USD has also been added by IEEE Power & Energy Society as a goodwill gesture.
- Granted 1000 USD Travel grant for IEEE PES Chair Meeting, However, due to visa delay, couldn't attended.
- Received two times 750 USD, 750 USD travel grant for attending IEEE PELS Chapter meeting, however due to short span of time unable to apply for visa.
- Received 1000 USD grant for industry student internship from IEEE PELS in 2026
- Received two time airfare from HEC, Pakistan for going and coming back from Pakistan to Malaysia and Malaysia to Pakistan during PHD perusal.
- 200 USD for IEEE PELS Day 2025 from IEEE PELS USA.

Speaker/ Workshops/ Lectures, invited Speaker, Academic Judge for the Final Year Design Projects and Students papers competitions

- Member, IEEE PELS (Power Electronics Society), ITRG stands for International Technology Roadmap of High-Power Electronics for Modern Energy Grids, **Grid services & Integration, Voltage control** – Mahesh Kumar. The working groups will review the latest state of the art and challenges with the industry practice of developing high (>10 MW) power electronic converters and systems.
- **Member National Technical Committee**, 8th International Conference on Energy, Environment, and Sustainable Development (EESD-2026), March 25 - 26, 2026 • MUET, Jamshoro
- **Conference Secretary**, 2nd International Conference on Emerging Technologies in Electronics, Computing and Communications, held in Mehran University of Eng. & Technology Jamshoro, Pakistan on 23-25 April 2025.
- **Tutorial Chair**, 8th International Multi-topic ICT Conference **IMTIC'25**, 29-31- December 2025 at Mehran University of Engineering and Technology Jamshoro and Sir Syed University of Engineering and Technology Karachi.
- **Session Chair**, 7th International Multi-topic ICT Conference **IMTIC'23**, 10-12- May 2023 at Sir Syed University of Engineering and Technology Karachi.
- **Session Chair**, 2nd International Conference on Emerging Technologies in Electronics, Computing and Communications (ICETECC 2025) held on 23-25 April 2025 at Mehran UET Jamshoro.
- **Session Chair**, 8th International Electrical Engineering Conference (8th IEEC-2023) on Friday the 25th & Saturday 26th August, 2023 at NED University of Engineering & Technology, Pakistan.
- **Keynote Speaker**, International Multi-disciplinary Conference in Emerging Research Trends (**IMCERT**), Karachi, Pakistan during January 4-5 2023.
- **Program Committee Member, Energy Informatics. Academy Conference 2025 (EIA 2025)** 3-6 December 2025, Universiti Tenaga Nasional (UNITEN), Kuala Lumpur, Malaysia.
- **Member National Technical Committee**, 7th International Conference on Energy, Environment, and Sustainable Development (EESD-2024), February 19-20, 2025 MUET, Jamshoro
- **Poster Presentation Committee**, 5th International Conference on Energy, Environment, and Sustainable Development (EESD-2024), November 14-15, 2018 MUET, Jamshoro
- **Program Committee Member, Global Conference on Wireless & Optical Technologies (GCWOT)** 25 -27 September 2024 Universidad de Málaga, Campus de Teatinos Malaga.

Professional Academic & Leadership Roles

Member of numerous high-level academic, research and administrative bodies at Mehran University.

- Served on **University Assessment Team** constituted by the Vice Chancellor, Mehran University of Engineering & Technology, for reviewing Self-Assessment Reports (SARs) of postgraduate programs in compliance with HEC QAA updated standards (PREE, 2026).
- Appointed as **Panel Member** for Review of Institutional Performance and Enhancement (RIPE) for the academic year 2025–2026, as notified by the Vice Chancellor, Mehran University of Engineering & Technology (MUET).
- **Nominated as Institutional Program Team (IPT) Member** for the 10th round of Charter Inspection & Evaluation Committee (CIEC), Sindh Higher Education Commission, as notified by the Vice Chancellor, Mehran University of Engineering & Technology (2025).
- Served as **Internal Member, Board of Studies (PhD)** for the Department of Electrical Engineering, MUET, contributing to curriculum development, research program oversight, and academic policy formulation for the 2024–2027 term, under notification of the Vice Chancellor.
- Appointed as **Committee Member for ME/MS/MPhil Program Review**, constituted by the Vice Chancellor, Mehran University of Engineering & Technology (MUET), in accordance with approved Terms of Reference (TORs) (2024).
- Served on University **Assessment Team** constituted by the Vice Chancellor, MUET, responsible for quality assessment of postgraduate and doctoral programs (ME/MS/MPhil & PhD) under HEC QAA Self-Assessment framework for the academic year 2021 - 2025.
- **Faculty Ambassador**, Business Incubation Center (BIC): Officially appointed by the Vice-Chancellor (2025–Present). Tasked with fostering an entrepreneurial culture, identifying student innovations, and acting as the primary departmental liaison to facilitate mentorship, funding pathways, and industry connections for startups.
- **Member Accreditation Committee**, Reviewing and preparing Self-Assessment Report for Accreditation of various courses at undergrad and post grad level in Mehran University..
- **Member Outcome-Based Education (OBE) System**, Preparing, assisting and reviewing departmental rubrics, evaluation procedure and templates to produce well-equipped EL graduates. Improved education quality and assurance of learning by closely working with faculty and Quality Enhancement Cell (QEC).
- **Member Departmental Course Review Committee**, Working closely with departmental faculty to see the undergrad courses fulfilling the needs of industry and academia

- **Member Board of Studies**, Key members for BOS for undergraduate and postgraduate courses at Dept. of Electrical Engineering at Mehran University.
- **Member Board of Faculty**, Member BOF, representing department of electrical engineering at Faculty meetings.
- **ISO Coordinator**, ISO coordinator with Department of Electrical Engineering at Mehran University.
- **Postgraduate Coordinator**, Postgraduate coordinator for Department of Electrical Engineering at Mehran University for year 2021-2025

Editorial Board Members for Academic Research Journals

1. Editorial Team Member: Member National Advisor for Institution of Electrical & Electronics Engineers Pakistan IEEEEP New Horizons journal
<http://www.ieeeepnhrij.org/index.php/ieeeep/about/editorialTeam>
2. Associate Editor: Journal of Applied Engineering & Technology (JAET)
<http://jae-tech.com/index.php/jaet/about/editorialTeam>

Accomplishments

- **PHD Scholarship**, Universiti Teknologi Petronas (UTP) Malaysia offered a fully funded PHD scholarship Graduate assistantship (GA).
- **Organized** 3-Days Online Training Session on PSS@SINCAL SIEMENS Software at Electrical Engineering, Mehran University with Yugal Kishore, Siemens Australia 17-19 October 2022.
- **Speaker** at IEEE PES Karachi Section Technical Seminar "Electrical Distribution System considering uncertainties" on Oct-2022
- **Attended**, IEEE Panel of Conference Organizers conference on "arranging the research conferences in collaboration with IEEE". The conference was arranged by POCO IEEE section Karachi at IoBM Karachi on 17 September 2022
- **Organized**, IEEE PES MUET Student branch Chapter session on Global UGRAD program 2023 with title "Global UGRAD 2023: How to prepare the best Application". Session was conducted by student Alumnus of Global UGRAD Spring 22 at Mehran University on 16 September 2022.
- **Organized Safety Drives**, Helmet Awareness Drive at Main University Jamshoro on the platform of BIG EVENT MUET on 11 August 2022.
- **Attended** Training as Dept. ISO Coordinator at Senate Hall of the MUET, Jamshoro in June 2022.

- **Organized**, "Role of Wind Energy for Powering Sustainable Future"- theme of event on IEEE POWER AND ENERGY SOCIETY (PES) day. Speaker Assadullah Qureshi (FFC), on 26 April 2022.
- **Speaker** for "Role of Wind Energy for Powering Sustainable Future" seminar at Mehran University of Engineering and Technology Pakistan on 26 April 2022.
- **Excellence Award** by IEEEEP Hyderabad Local Centre at Mehran University, Jamshoro at a one day session titled "MEET THE PERSONALITY" at Senate Hall of the University.
- **Speaker**, Uplift your career with IEEE PES, 15th IEEE Pakistan Students, Young Professionals, Women in Engineering, SIGHT Congress at Hamdard University Karachi 28 – 29 December 2022.
- **Organized** for Tutorial Session on Innovations in European Union Projects on Energy for EESD 5th international Conference at Mehran University Jamshoro, Pakistan Nov 2018.
- **Participated** in SEDEX 37 post graduate research exhibition on 3-4 August 2016 at Universiti Teknologi PETRONAS, Malaysia.
- Completed one-day On-Grid Solar PV Trainer training program on 6th April 2016 at Universiti Teknologi Petronas, Malaysia.
- Completed Research Methodology Course during PHD studies at Universiti Teknologi Petronas, Malaysia on 5th January 2015.

Other Services and Contributions

a. Reviewer for Projects

- Reviewed more than a 10 national research projects under research and development division, Higher Education Commission Pakistan.
- Reviewer for the renowned journals: Journal of International Transactions on Electrical Energy Systems, Wiley. International Journal of Electrical Power & Energy Systems, Electric Power Components and Systems, International Journal of Renewable Energy Research, MDPI-Energies, Sustainability, Electronics, Symmetry, Applied sciences.
- Reviewer for conferences
- Academic reviewers, evaluators, and advisors for PhD Scholar at Bahria University

b. External Examiner

- Serving as a PhD and Master thesis examiner for several renowned universities in Pakistan. i.e. Quaid e Awam Univeristy Nawabshah, Bahria Univeristy Karachi.
- Serving as External Examiner for Final Year Project viva-voce at Quaid e Awam University Nawabshah, Pakistan.

Potential Research

- a. Uncertainty, inequity, and external disturbance are common problems in a conventional grid integrated WECS. Hence, 3L-NPC is chosen for grid integrated DFIG for intermediate voltage and high-power applications. The voltage and current waveforms of the 3L-NPC back-to-back converter with no load and with auxiliary load have been analysed and discussed. Besides, an analytical model of a nonlinear controller based on MSTFOTSMC for RSC and GSC control and stabilisation was combined with grid integrated DFIG. During three phase faults, STATCOM is used with the proposed controller to improve the power quality of grid integrated DFIG. Besides, Harmonic Analysis of Grid Integrated DFIG has been done under three phase faults. The results suggests that the proposed controller with STATCOM shows better performance for improving voltage profile and reducing harmonics distortion than that of proposed controller without STATCOM.

PhD student (Graduated)

1. **Mansoor Soomro (16Ph.DEnSE 05): development of controller design for power quality improvement of grid integrated wind energy conversion system.**
Mr. Mansoor is granted 2.0 Million Pkr grant for pursuing his Ph.D studies under the endowment scholarship at Mehran university of Engineering and Technology Jamshoro, Pakistan
- b. It is seen that material-based hydrogen storage provides several advantages over physical hydrogen storage system due to fact that material-based hydrogen storage process does not require high temperature and pressure. This study employed the h-BN monolayer system for storing the hydrogen molecules due to having extraordinary properties. h-BN in pure state is not suitable for hydrogen store process. The addition of dopant can effectively create extra charge on the surface of 2D monolayer systems, which can be used for adsorption of the hydrogen molecules. This research uses DFT (Density Functional Theory) technique to examine the H₂ storage on Li-decorated h-BN monolayer. Moreover, the H₂ desorption simulations are achieved via the ab initio molecular dynamics (MD). The computed desorption temperature TD is 306 °K which is a suitable operating temperature. Hence this research demonstrates that Li-doped h-BN is a thermally stable and viable hydrogen storage material for hydrogen storage systems.

PhD student (Graduated)

1. **Ms. Kaneez Fatima (18IICTPhD28) Computational Investigations for Clean and Renewable Hydrogen Energy; its Storage and Applications Based on First-Principles Calculations**
Ms. Kaneez working on Approved project funded by Pakistan science foundation (PSF) titled "Computational investigations on Novel 2D Mono atomic hexagonal boron nitrides (h-BN) and transition metal nitrides (TMN) materials for hydrogen storage application" letter No. PSF/Res/S-MUET/Engg/(187) dated May2021.
- c. My recent projects focus on solving **practical power system and energy industry challenges**, particularly in **distribution network optimization, renewable energy integration, electric vehicle charging infrastructure, and smart grid enhancement**. These projects include **loss minimization in radial distribution**

systems through optimal placement and sizing of distributed generation (DG) and shunt capacitors using advanced meta-heuristic techniques (PSO, ANN-based methods), enhancement of PV hosting capacity in low-voltage networks, mitigation of voltage sag using dynamic voltage restorers, and improvement of short-circuit levels and power transfer capability through strategic system planning and FACTS implementation. Additional work involves techno-economic design of isolated and grid-connected microgrids, hybrid PV-hydrogen and renewable-based EV charging stations, biomass energy systems, industrial self-consumption solar with battery storage, and net-metering and carbon credit analysis frameworks to support sustainable energy transition.

List of Masters in Electrical Engineering Students Supervision:

S.#	Year	Name and Roll No.	Research Title	Supervisor	Co-Supervisor	Status
1	2026	Mr. Muhammad (23MEELP 26)	"PSO-PID Based Dynamic Voltage Restorer for Voltage Sag Compensation in the Distribution System'.	Dr. Mahesh Kumar	Prof. Dr. Aamir Mahmood Soomro	In progress
2	2026	Mr. Anees Qadir (23MEELP 14)	"Optimal Planning of Distribution Network Considering Multi-Scenario Load Models."	Dr. Mahesh Kumar	Prof. Dr. Aamir Mahmood Soomro	In progress
3	2026	Mr. Amaz Mushtaque (23MEELP 25)	"ANN Based Dynamic Voltage Restorer with Super Capacitor for Power Quality Enhancement in Distribution Network.	Prof. Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar	In progress
4	2026	Mr. Shafqat Hussain (23ME ELP 24)	"On grid Solar System and Carbon Credit Mechanism"	Dr. Mahesh Kumar	Prof. Dr. Aamir Mahmood Soomro	In progress
5	2026	Mr. Haseeb Hassan Shah (22MEELP 11)	"Economic Assessment of an Grid PV with Self Consumption and Battery Storage for Large Industrial loads."	Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar	Graduated
6	2025	Mr. Muzamil Ahmed, (21MEELP34)	Multi scenario Analysis of Renewable Energy	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro	Graduated

			Resource Based Off-Grid EV Charging Station.			
7	2025	Mr. Aneel Kumar, (23ME-ELP018),	Optimal Planning of Multi - Energy Distribution System for Electric Vehicle Charging Stations Using Meta-Heuristic Optimization Algorithm.	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro	Graduated
8	2025	Mr. Aaqib Fakhree-e-Aalam, (21MEELP15)	Implementation Of Net Metering Arrangement and Economic Analysis of Solar Power System at an HEL.	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro	Graduated
9	2025	Mr. Tufail Ahmed, (21 ME ELP 21)	Design of Effective Facts Controller on 500 KV Jamshoro-Dadu Transmission Line to Increase Power Transfer Capabilities and its Financial Impact.	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro	Graduated
10	2024	Ms. Noor Jahan (20MEELP25)	Techno-Economic Assessment Of Biomass Energy to Mitigate Power Shortage of Hyderabad Sindh	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro	Graduated
11	2024	Mr. JUNAID ABBASI (20MELP28)	Design and Techno-Economic Evaluation of DC Micro Grid for Rural Health Care System	Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar	Graduated
12	2024	Mr. Kamlesh Kumar (20MEELP34)	Slef-consumption and cost analysis of on-grid solar photovoltaic system for and industrial load	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro	Graduated
13	2023	Mr. Masroor Khan (18MEELP34)	Enhancing Photovoltaic Hosting Capacity in Low voltage Distribution Network.	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro	Graduated
14	2023	Mr. Asad Shaikh (20MEELP61)	Plug-in electric vehicle (PEV) Charging station based on hybrid PV-Hydrogen energy system.	Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar	Graduated

15	2023	Ms. Hafsa Shaikh (19MEELP41)	Optimal placement strategy of energy storage system in radial distribution network	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro	Graduated
16	2021	Mr. Waleed Ahmed (17 ME ELP 45):	Design of Isolated Micro Grid Using Renewable Energy Resources with Energy Storage Systems via HOMER Software (A Case Study of a Thar Community)	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro	Graduated
17	2023	Mr. Furqan Latif Memon (19 ME ELP 48)	Optimal Integration of Distributed Generation In Distribution System to Improve Short Circuit Current Level	Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar	Graduated
18	2020	Mr. Shezor Ahmed (18MEELP28))	Optimal Placement of Shunt Capacitor in Distribution System by Particle Swarm Optimization.	Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar	Graduated
19	2020	Mr. Faizan Channa (18MEELP-33)	Strategically plug-in electrical vehicle load management in radial distribution system.	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro	Graduated
20	2020	Mr. Owias Ahmed Memon	Optimal Placement of Shunt Capacitor in Distribution System by Particle Swarm Optimization.	Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar	Graduated
21	2020	Mr. Manzoor Hussain (18 ME ELP 11)	Optimal placement and sizing of DG in Radial Distribution System using PSO Technique	Dr. Mahesh Kumar	Prof. Dr. Ashfaq Ahmed Hashmani	Graduated
22	2019	Mr. Shahbaz Ahmed (17 ME ELP 47)	Optimal Siting and Sizing of DG in a HESCO Feeder Using Particle Swarm Optimization	Dr. Mahesh Kumar	Dr. Anwar Ali Sahito	Graduated

List of Undergraduate Students Supervision:

S.#	Roll No.	Title	Supervisor	Co-Supervisor
1	22EL02 22EL011 22-21EL123	Techno-Economic Analsis of PV Battery for Remote Electrical Vehicle Charging Station.	Dr. Mahesh Kumar	Prof. Dr. Aamir Mahmood Soomro
2	22EL069 22EL061 22-21EL128	Condition Monitoring of Wind Turbine and Optimizing their Performance and Efficiency	Dr. Mahesh Kumar	Dr. Mansoor Ahmed Soomro
3	22EL057 22EL007 22EL071	To Automated PV Impact and Net Load Forecasting for Grid Planning.	Prof. Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar
4	21EL043 21EL017 21EL063	Development an intelligent Energy Management System for micro grids using Microcontroller	Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar
5	20EL022 20EL040 20EL125	Improve the thermal efficiency of PV cell using cooling techniques	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro
6	20EL049 20EL012 20EL030	Increment in solar energy efficiency using thermal storage system	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro
7	19EL09 19EL10 19EL103	Anti- Islanding Protection for Solar Connected Grid System	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro
8	19EL56 19EL57 19EL58	Underground cable fault detector based on arduino GSM AND GPS	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro
9	19-18EL09 19EL79 19EL137	Smart health monitoring and protection system for distribution transformer	Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar
10	18EL75 18EL83 18EL183	Improved maximum power point tracking algorithms for Photovoltaic system	Dr. Aamir Mahmood Soomro	Dr. Mahesh Kumar
11	18EL56 18EL47 18EL131	Intelligent control of BESS and PV in LV distribution system	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro
12	17EL58 17EL57 17EL54	Design and Simulation Analysis of Hybrid Microgrid using Solar and Wind for Rural Population.	Dr. Mahesh Kumar	Dr. Aamir Mahmood Soomro

Research Publications

Journal Publications:

1. Khalil, Muhammad Waqas, Syed Ali Abbas Kazmi, Mustafa Anwar, **Mahesh Kumar Rathi**, Fahim Ahmed Ibupoto, and Mukesh Kumar Maheshwari. "A Collaborative Energy Management and Price Prediction Framework for Multi-Microgrid Aggregated Virtual Power Plants." *Sustainability* (2071-1050) 18, no. 1 (2026). (IF = 3.3) <https://doi.org/10.3390/su18010275>
2. Mansoor Soomro, Zubair Ahmed Memon, **Mahesh Kumar**, Mazhar Hussain Baloch, Nayyar Hussain Mi, Laveet Kumar, A review of single stage and multistage power converters for doubly fed induction generator integrated with power grid. *Wind Engineering*. 2024;49(2):513-535. doi:10.1177/0309524X241263764, (IF = 1.8)
3. **Mahesh Kumar**; Amir Mahmood Soomro; Syed Ali Abbas Kazmi; Aneel Kumar Techno-Economic Comparative Analysis of an Off-Grid PV-Wind-Hydrogen Based EV Charging Station under Four Climatically Distinct Cities in Pakistan *International Journal of Hydrogen Energy*, Elsevier, 93, pp.1268-1282 (2024) (I.F = 8.1) <https://doi.org/10.1016/j.ijhydene.2024.11.074>.
4. **Mahesh Kumar**, Aneel Kumar, Amir Mahmood Soomro, Dr Mazhar Baloch, Dr. Sohaib Tahir, Muzamil Ahmed Shaikh A Comprehensive Review of Optimizing Multi-Energy Multi-Objective Distribution System with Electric Vehicle Charging Stations, *World Electric Vehicle Journal*, 15, no. 11: 523, MDPI (2024). <https://doi.org/10.3390/wevj15110523> (I.F = 2.6).
5. Amir Mahmood Soomro, **Mahesh Kumar Rathi**, Asad Shaikh, Laveet Kumar, Mamdouh El Haj Assad, M.A.Ehyaei, Techno-Economic Analysis Of Stand-Alone Hybrid PV-Hydrogen-Based Plug-In Electric Vehicle Charging Station, *Energy Report*, Elsevier (2024) (I.F = 4.7), <https://doi.org/10.1016/j.egyr.2024.09.019>
6. Noor Saleem Khan, Syed Ali Abbas Kazmi *, Mustafa Anwar *, Saqib Ur Rehman Mughal, Kafait Ullah, **Mahesh Kumar Rathi**, Ahmad Salal Enhancing renewable energy integration in developing countries: A policy-oriented analysis of net-metering in Pakistan amid economic challenges, *Energies* (2024): (I.F = 3.37) <https://doi.org/10.3390/su16146034>
7. Mansoor Soomro, Zubair Ahmed Memon, **Mahesh Kumar**, Mazhar Hussain Baloch, Nayyar Hussain Mirjat, Laveet Kumar, A Comprehensive Review of Single-Stage and Multistage Converters for Doubly Fed Induction Generators (DFIG) Integrated With Grid, *Energies*, (I.F = 1.5) <https://doi.org/10.1177/0309524X241263764>
8. Kumar, Aneel, **Mahesh Kumar**, Amir Mahmood Soomro, and Laveet Kumar. "Techno-Economic Optimization of Novel Stand-Alone Renewable Based Electric Vehicle Charging Station near Bahria Town Karachi, Sindh Pakistan." *Energy Engineering* 121, no. 6 (2024), **Scopus indexed**, <https://doi.org/10.32604/ee.2024.049977>
9. Kamlesh Kumar, Amir Mahmood Soomro, **Mahesh Kumar**, Laveet Kumar, Müslüm Arıcı, "Sustainable On-Grid Solar Photovoltaic Feasibility Assessment for Industrial Load", *Journal of Central South University*, (I.F = 3.2) <https://doi.org/10.1007/s11771-023-5488-0>
10. Abdalla, Elnazeer Ali Hamid, **Mahesh Kumar**, Izzeldin Idris Abdalla, Salah Eldeen Gasim Mohamed, Amir Mahmood Soomro, Muhammad Irfan, Saifur Rahman, and Grzegorz

- Nowakowski. "Modeling and Optimization of Isolated Combined Heat and Power Microgrid for Managing Universiti Teknologi PETRONAS Energy." IEEE Access (2023). (I.F = 3.54) <http://doi.org/10.1109/ACCESS.2023.3296428>
11. Fatima, Kaneez, Muhammad Rafique, Amir Mahmood Soomro, and **Mahesh Kumar**. "Tailoring Hydrogen Adsorption and Desorption Properties of Li doped SV Monolayer h-BN Systems using Ab-Initio Calculations." Canadian Journal of Physics ja (2023). (I.F = 1.358) <https://cdnsiencepub.com/doi/abs/10.1139/cjp-2023-0072>
 12. Leghari, Zohaib Hussain, Mohammad Yusri Hassan, Dalila Mat Said, Laveet Kumar, **Mahesh Kumar**, Quynh T. Tran, and Eleonora Riva Sanseverino. "Effective Utilization of Distributed Power Sources under Power Mismatch Conditions in Islanded Distribution Networks." Energies 16, no. 6 (2023): 2659. (I.F = 3.252) <https://doi.org/10.3390/en16062659>
 13. Leghari, Zohaib Hussain, **Mahesh Kumar**, Pervez Hameed Shaikh, Laveet Kumar, and Quynh T. Tran. "A Critical Review of Optimization Strategies for Simultaneous Integration of Distributed Generation and Capacitor Banks in Power Distribution Networks." Energies 15, no. 21 (2022): 8258. (I.F = 3.252) <https://doi.org/10.3390/en15218258>
 14. **Mahesh Kumar**, Amir Mahmood Soomro, Waqar Uddin, and Laveet Kumar. "Optimal Multi-Objective Placement and Sizing of Distributed Generation in Distribution System: A Comprehensive Review." Energies 15, no. 21 (2022): 7850. (I.F = 3.252) <https://doi.org/10.3390/en15217850>
 15. Fatima, Kaneez, Amir Mahmood Soomro, Muhammad Rafique, and **Mahesh Kumar**. "Hydrogen storage on flat land materials, opportunities, and challenges: A review study." Journal of the Chinese Chemical Society 69, no. 4 (2022): 663-680. <https://doi.org/10.1002/jccs.202200039> (I.F = 1.753)
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 17. Hafsa Imran Sheikh, **Mahesh Kumar Rathi**, and Amir Mahmood Soomro. "Optimal integration of battery energy-storage system with high penetration of renewable energy in radial distribution network." Clean Energy 6, no. 3 (2022): 404-411 <https://doi.org/10.1093/ce/zkac027> A . (I.F = 1.33)
 18. Mansoor Ahmed Soomro, Zubair Ahmed Memon, **Mahesh Kumar**, Mazhar Hussain Baloch "Wind energy integration: Dynamic modeling and control of DFIG based on super twisting fractional order terminal sliding mode controller" Energy Report <https://doi.org/10.1016/j.egy.2021.09.022> (I.F = 6.87)
 19. Kumar, Kamlesh, **Mahesh Kumar**, and Amir Mahmood Soomro. "To Design an off Grid PV System for an electrified area of District Tharparkar, Pakistan." International Journal of Green Energy 18.9 (2021): 920-932. <https://doi.org/10.1080/15435075.2021.1884867> (I.F = 2.73)

20. **Mahesh Kumar**, Waleed Memon, Amir Mahmood Soomro "The Design of Isolated Micro Grid Using Renewable Energy Resources with Energy Storage System via HOMER Software A Case Study of a Thar Community." Sukkur IBA Journal of Computing and Mathematical Sciences 5.1 (2021): 53-58.
21. Muhammad Zeeshan Malik, **Mahesh Kumar**, Amir Mahmood Soomro, Mazhar Hussain Baloch, Mehr Gul, Muhammad Farhan, Ghulam Sarwar Kaloi Strategic planning of renewable distributed generation in radial distribution system using advanced MOPSO method, Energy Reports, Volume 6, November 2020, Pages 2872-2886, <https://doi.org/10.1016/j.egyr.2020.10.002>. [PDF](#) (I.F = 6.87)
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30. Saindad Mahesar, Ghulam S. Kaloi, **Mahesh Kumar**, Power Management of a Stand-Alone Hybrid (Wind/Solar/Battery) Energy System: An Experimental Investigation *International Journal of Advanced Computer Science and Applications*, Vol. 9, No. 6, 2018 10/10/2018. <https://doi.org/10.14569/IJACSA.2018.090631>. **PDF**
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32. **Mahesh Kumar**, Amir Mahmood Soomro, Abdul Hakeem Memon, Perumal Nallagownden Irraivan Elamvazuthi, Bhagwan Das Particle Swarm Optimization Technique Based Power Loss Reduction in Radial Distribution System *IJCSNS International Journal of Computer Science and Network Security*, VOL.18 No.12, December 2018. 30/12/2018 **PDF ISI indexed**
33. B. Das , **M. Kumar** , S.Ali , A. F. Chandio , N. A. S. T. Pratomo , M. F. L. Abdullah, Performance Analysis of BPSK-OFDM and DQPSK-OFDM Schemes for Wireless Communication System, *Technical Journal, University of Engineering and Technology (UET) Taxila, Pakistan* Vol. 23 No. 1-2018 ISSN:1813-1786 (Print) 2313-7770 (Online) 01/09/2018 **PDF**
34. **K. Mahesh**, P. Nallagownden, and I. Elamvazuthi, "Optimal Placement and Sizing of Distributed Generators for Voltage-Dependent Load Model in Radial Distribution System" *Renewable Energy Focus (IF 4.8)*, Volume 19–20, Number 00, June 2017, **Scopus indexed**. 01/06/2017. <https://doi.org/10.1016/j.ref.2017.05.003>. **PDF**
35. **K. Mahesh**, P. Nallagownden, and I. Elamvazuthi, "Optimal Configuration of intermittent and non-intermittent Type of Renewable Energy Sources and Capacitor banks into Radial Distribution System," *Energies*, vol.10, June 2017, **ISI/Scopus indexed with 2.676 impact factor (online)**. 14/06/2017 <https://doi.org/10.3390/en10060811>. **PDF**
36. **K. Mahesh**, P. Nallagownden, and I. Elamvazuthi, "Advanced Pareto Front Non-Dominated Sorting Multi-Objective Particle Swarm Optimization for Optimal Placement and Sizing of Distributed Generation," *Energies*, vol. 9, DEC 2016, **Q2 ISI/Scopus indexed with 2.262 impact factor (online)**. 25/11/2016. <https://doi.org/10.3390/en9120982>. **PDF**
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38. **Mahesh Kumar**, Bhagwan Das, Mazhar Hussain Baloch, Perumal Nallagownden, Irraivan Elamvazuthi, Abid Ali, Optimal Placement and Sizing of Distributed Generators and Distributed-Static Compensator in Radial Distribution System: Distributed Generators and Distributed Static Compensator International Journal of Energy Optimization and Engineering Volume 8 • Issue 1 • January-March 2019. 01/10/2018. <https://doi.org/10.4018/IJEOE.2019010103>. [PDF](#) [ISI indexed](#)
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40. Ghulam Sarwar, Mazhar Hussain, **Mahesh Kumar**, Rasool BuxLashari,(2014) Analysis and Estimation of Technical Losses in Urban Distribution Feeders Bahawalpur, Sindh University Research Journal (science series) Vol. 46 (1) p.p. 33-36. 03/03/2014 [PDF](#)
41. M.H. Baloch, **M.K. Maheshwari**, A.AMemon (2013) Energy conservation through energy efficient motors in Pakistan's industrial sector, Sindh University Research Journal (science series) Vol. 45 (3) p.p. 501-505. 18/08/2013 [PDF](#)
42. **Mahesh Rathi**, Faheemullah Shaikh, Pervez Hameed Shaikh, M. Aslam Uqailli "Energy Conservation through Motors in Pakistan, Industrial Sector- Need to use of Energy Efficient Motors. *International Journal of Computer Applications* 54(5):1-7, September 2012. Published by Foundation of Computer Science, New York, USA. 09/09/2012 <https://doi.org/10.5120/8559-2141> [PDF](#)

Conference Proceedings:

43. M. Kumar and A. M. Soomro, "Artificial Intelligence Based Optimal Integration of Solar Photovoltaic in Electric Power Distribution Network," 2024 Global Conference on Wireless and Optical Technologies (GCWOT), Malaga, Spain, 2024, pp. 1-5, doi: 10.1109/GCWOT63882.2024.10805583.
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46. **K. Mahesh**, P. Nallagownden, and I. Elamvazuthi, "Optimal Configuration of DG in Distribution System: An Overview," in MATEC Web of Conferences, 2016, p. 01007, **Scopus indexed (online)**. 13/11/2015. <https://doi.org/10.1051/mateconf/20163801007>.
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Book Chapter

50. **K. Mahesh**, P. Nallagownden, and I. Elamvazuthi, "Optimal Placement and Sizing of Distributed Generators for Electrical Distribution System Using Metaheuristic Optimization Algorithm: Swarm Intelligence Based Distributed Generation", "Emergent Research on the Application of Optimization Algorithms", IGI Publisher, 2017, **Online**. 06/07/2017 <https://doi.org/10.4018/978-1-5225-2990-3.ch021>. [PDF](#)
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Key Research Projects

1. Novel Controller Design for Grid Connected Wind Turbines.

The uncertain and fluctuating nature of wind flow cause power quality problems. Therefore a model has been designed and response of power quality issues like voltage sag and harmonic distortion is analysed both at no load, with load and under faults at the point of common coupling (PCC). Considering, high penetration of wind energy conversion system (WECS) a voltage and frequency regulation remains a challenge with respect to faults, therefore, both asymmetrical and symmetrical faults at PCC require careful analyses. Symmetrical faults are most intense inflicting the stator winding to low voltage disturbing low voltage ride through (LVRT) functionality of DFIG. The vector control strategy using proportional integral (PI) controllers has been applied to control DFIG parameters of both rotor side converter (RSC) and grid side converter (GSC). In addition, shunt transformer interfaced series grid side converter is used as an auxiliary hardware circuitry for improving the performance of DFIG under fault symmetrical fault conditions. The MATLAB simulation results demonstrate that the system effectively controls fluctuations in voltage and current thereby improving the power quality in unbalanced situations. However, uncertainty, inequity, and external disturbance are common problems in a conventional grid integrated WECS. Hence, 3L-NPC is chosen for grid integrated DFIG for intermediate voltage and high-power applications. The voltage and current waveforms of the 3L-NPC back-to-back converter with no load and with auxiliary load have been analysed and discussed. Besides, an analytical model of a nonlinear controller based on MSTFOTSMC for RSC and GSC control and stabilization was combined with grid integrated DFIG. During three phase faults, STATCOM is used with the proposed controller to improve the power quality of grid integrated DFIG. Besides, Harmonic Analysis of Grid Integrated DFIG has been done under three phase faults.

2. Artificial Intelligence Based Optimal Integration of Renewable Distributed Energy Resources in Distribution Network:

a. Renewable Distributed Energy Resources in Distribution Network using advanced Multi-objective Particle Swarm Optimization Method:

The integration of wind turbine, solar photovoltaic based distributed generation in low voltage distribution network has many advantages. However, the output power from wind turbine and solar PVs are intermittent in nature. The production of these intermittent power generations may introduce many challenges in distribution networks. Therefore, a strategic optimal planning with the help of multi-objective optimization have

been proposed. The intermittency of wind speed and solar irradiance are handled with multi-state modeling using suitable probability distribution functions. The three objective functions, i.e., power loss reduction, voltage stability improvement, and voltage deviation minimization are optimized using advanced Pareto-front non-dominated sorting multi-objective particle swarm optimization method. First a set of non-dominated Pareto-front data are called from the algorithm. Later, a fuzzy decision technique is applied to extract the trade-off solution set. The effectiveness of the proposed methodology is tested on the standard IEEE 12, IEEE 33, IEEE 69, HESCO, Pakistan distribution feeder and Portuguese distribution feeders. The overall results reveal that combination of renewable distributed generations and capacitor banks are dominant in power loss reduction, voltage stability and voltage profile improvement.

- b. **Voltage dependent load models in distribution network:** The electricity demand is increasing every year and there are many customers and their power requirement, therefore, different load models considering, residential, industrial, commercial and mix load or CP, CI, CZ or ZIP load model have been test with base year and next 5 years using advanced MOPSO optimization.
- c. **Optimal integration of battery energy-storage system with high penetration of renewable energy in radial distribution network:**

Considering the intermittent nature of renewable energy, a storage system to reserve power in off-peak hours and then to supply it during peak hours is necessary. However, if these storage devices in a network are not placed in an appropriate manner, the advantages can never be achieved. In this study, the allocation and sizing strategies of a battery energy-storage system (BESS) in an optimal way are proposed to improve the performance of the radial distribution networks. The test network adopted is a standard IEEE 33 bus network that is integrated with solar power. Simulations are carried out in DIgSILENT PowerFactory for the distribution network model and the process of optimization is done through MATLAB®. The optimization algorithm adopted is the genetic algorithm. The system is studied for 24 hours with a step size of 15 minutes, employing a time-sweep analysis of dynamic load. It is studied under three different scenarios and the results are compared. The results illustrate a considerable reduction in power losses in the case of the optimized BESS. The total losses before the optimization process are 16 365.57 kW. The total losses after the employment of the proposed solution are 10 246.5 kW, which means that the losses are reduced by greater than 62%.

3. Autonomous Distribution Network for Power

The integration of distributed generation (DG) into a power distribution network allows the establishment of a microgrid (MG) system when the main grid experiences a malfunction or is

undergoing maintenance. In this case, the power-generating capacity of distributed generators may be less than the load demand. This study presents a strategy for the effective utilization of deployed active and reactive power sources under power mismatch conditions in the islanded distribution networks. Initially, the DGs' and capacitors' optimal placement and capacity were identified using the Jaya algorithm (JA) with the aim to reduce power losses in the grid-connected mode. Later, the DG and capacitor combination's optimal power factor was determined to withstand the islanded distribution network's highest possible power demand in the event of a power mismatch. To assess the optimal value of the DG-capacitor pair's operating power factor ($p_{fsource}$) for the islanded operation, an analytical approach has been proposed that determines the best trade-off between power losses and the under-utilization of accessible generation. The test results on 33-bus and 69-bus IEEE distribution networks demonstrate that holding the islanded network's load power factor (p_{fload}) equal to $p_{fsource}$ during the power imbalance conditions allows the installed distributed sources to effectively operate at full capacity. As expected, the proposed strategy will assist the utility companies in designing efficient energy management or load shedding schemes to effectively cope with the power mismatch conditions.

4. Techno-Economic Optimization of Novel Stand-Alone Renewable Based Electric Vehicle Charging Station:

Electric vehicles (EVs) are the most interesting and innovative technology in the 21st century because of their enormous advantages, both technically and economically. If Electric vehicles demand increases drastically then conventional power stations will not bear their demand and if they generate electricity by conventional means it will be very costly and may further add greenhouse gases. Therefore, our research provides the techno-economic assessment of a stand-alone renewable-dependent electric vehicle charging station, we used HOMER Pro software for techno-economic assessment. A hybrid system comprising solar Photovoltaic/Wind Turbine/Fuel cells and battery storage is included in the model.

In another study, Hybrid PV-hydrogen and PV-hydrogen-battery are the two different scenarios that are carried out and compared based on their both technical as well as financial standpoints. The simulation results are evident that the hybrid PV- hydrogen-battery energy system has much more financial and economic benefits as compared with the PV-hydrogen energy system.

5. Computational investigations on Novel 2D Mono atomic hexagonal boron nitrides (h-BN) and transition metal nitrides (TMN) materials for hydrogen storage application:

It is seen that material-based hydrogen storage provides several advantages over physical hydrogen storage system due to fact that material-based hydrogen storage process does not require high temperature and pressure. Our study employed the h-BN monolayer system for storing the hydrogen molecules due to having extraordinary properties. h-BN in pure state is not

suitable for hydrogen store process. The addition of dopant can effectively create extra charge on the surface of 2D monolayer systems, which can be used for adsorption of the hydrogen molecules. we uses DFT (Density Functional Theory) technique to examine the H₂ storage on Li-decorated h-BN monolayer. Moreover, the H₂ desorption simulations are achieved via the ab initio molecular dynamics (MD). The computed desorption temperature TD is 306 °K which is a suitable operating temperature. Hence our research demonstrates that Li-doped h-BN is a thermally stable and viable hydrogen storage material for hydrogen storage systems.